

MH2500 PROBABILITY AND INTRODUCTION TO STATISTICS
NANYANG TECHNOLOGICAL UNIVERSITY

Tutorial 5

16/09/2021

.....

We will discuss several questions from Chapter 5 in Ross' book.

Tutorial Questions

- (1) Consider n coins, each of which independently comes up heads with probability p . Suppose that n is large and p is small, and let $\lambda = np$. Suppose that all n coins are tossed; if at least one comes up heads, the experiment ends; if not, we again toss all n coins, and so on. That is, we stop the first time that at least one of the n coins come up heads.

- Let X denote the total number of heads that appear.

Which of the following reasonings concerned with approximating $P\{X = 1\}$ is correct (in all cases, Y is a Poisson random variable with parameter λ)?

- (a) Because the total number of heads that occur when all n coins are rolled is approximately a Poisson random variable with parameter λ ,

$$P\{X = 1\} \approx P\{Y = 1\} = \lambda e^{-\lambda}.$$

- (b) Because the total number of heads that occur when all n coins are rolled is approximately a Poisson random variable with parameter λ , and because we stop only when this number is positive,

$$P\{X = 1\} \approx P\{Y = 1 | Y > 0\} = \frac{\lambda e^{-\lambda}}{1 - e^{-\lambda}}.$$

- (c) Because at least one coin comes up heads, X will equal 1 if none of the other $n - 1$ coins comes up heads. Because the number of heads resulting from these $n - 1$ coins is approximately Poisson with mean $(n - 1)p \approx \lambda$,

$$P\{X = 1\} \approx P\{Y = 0\} = e^{-\lambda}.$$

- (2) Consider the function

$$f(x) = \begin{cases} C(2x - x^3), & 0 < x < \frac{5}{2} \\ 0, & \text{otherwise} \end{cases}$$

Could f be a probability density function? If so, determine C . How about $g(x)$ given by

$$g(x) = \begin{cases} C(2x - x^2), & 0 < x < \frac{5}{2} \\ 0, & \text{otherwise} \end{cases}$$

- (3) The probability density function of X is given by

$$f(x) = \begin{cases} a + bx^2, & 0 \leq x \leq 1 \\ 0, & \text{otherwise.} \end{cases}$$

If $\mathbb{E}(X) = 3/5$, find a and b .

- (4) The probability density function of X , the lifetime of a certain type of electronic device (measured in hours), is given by

$$f(x) = \begin{cases} \frac{10}{x^2}, & \text{if } x > 10; \\ 0, & \text{if } x \leq 10. \end{cases}$$

- (a) Find $P\{X > 20\}$.
 - (b) What is the cumulative distribution function of X ?
 - (c) What is the probability that of 6 such types of devices at least 3 will function for at least 15 hours? What assumptions are you making?
- (5) Let $X \sim N(3, 16)$. Find $P\{X > 0\}$.
- (6) Stanford accepts 2480 students and each student has a 68% chance of attending. Let X be the number of students who will attend. Then $X \sim \text{Bin}(2480, 0.68)$. What is $P(X > 1745)$?
- (7) If X is an exponential random variable with parameter λ , and $c > 0$, show that cX is exponential with parameter λ/c .

Assignment Questions

- (1) If Y is uniformly distributed over $(0, 5)$, what is the probability that the roots of the equation $4x^2 + 4xY + Y + 2 = 0$ are both real?
- (2) One thousand independent rolls of a fair die will be made. Compute an approximation to the probability that the number 6 will appear between 150 and 200 times inclusively. If the number 6 appears exactly 200 times, find the probability that the number 5 will appear less than 150 times.
- (3) Suppose that X is a normal random variable with mean 5. If $P\{X > 9\} = 0.2$, approximately what is $\text{Var}(X)$?
- (4) Let Z be a standard normal random variable, and let g be a differentiable function with derivative g' .
 - (a) Show that $\mathbb{E}(g'(Z)) = \mathbb{E}(Zg(Z))$, assuming that $g(x)/e^{x^2/2} \rightarrow 0$ as $|x| \rightarrow \infty$.
 - (b) Show that $\mathbb{E}(Z^{n+1}) = n \mathbb{E}(Z^{n-1})$.
 - (c) Find $\mathbb{E}(Z^4)$.
- (5) Suppose that the length of a phone call in minutes is an exponential random variable with parameter $\lambda = 1/10$. If someone arrives immediately ahead of you at a public telephone booth, find the probability that you will have to wait
 - (a) more than 10 minutes;
 - (b) between 10 and 20 minutes.